Architectural Proposal for Evaluation Agent

Literature Review

Byun et al. (2025) show that SOTA (state of the art) LLMs like GPT-4o achieve strong correlation (up to 0.98) with human graders and upto 55% exact score agreement. However, some variability remains in technical and open ended responses.
This work indicates the suitability of LLMs for ASAG (automated short answer grading) tasks, which form the majority of the answer types featured in Indian examinations.

Xu et al. (2025) propose Chain-of-Thought prompting to enhance essay scoring by introducing a rubric alignment. Their goal is to make automated essay scoring more explainable, thus making justification of scoring easier and more defensible in face of challenges from concerned students, educators and parents. However, their utilization of RLHF as well as finetuned models limit applicability outside their original domain of IELTS essay grading.
However, the key idea that can be taken away from this paper is decomposing a golden key into rubrics, that are then matched against the student's answer in a deterministic fashion before incorporating deeper semantic review using the LLM.

Proposed Architecture

From the two papers, a possible architecture for the Evaluation agent is:
Part 1: A crossencoder model that compares two sequences (here, a rubric sequence and the student answer sequence) at a very deep (token) level and outputs a similarity score.
Part 2: An LLM that then takes the similarity scores outputted by the crossencoders and uses that as the basis to:

- Generate improvement suggestions for the student
- Use those as the basis for a reasoning-driven mark assignment

This architecture combines the deterministic nature of similarity matching while allowing for semantic interpretations that can better handle the uncertain and imprecise language used by students.

The key to this approach is the Golden Key. The best way to precompute this Golden Key is to ask the end user - ie, the educator - to input specific points/metrics that they want to see in the student answer.
For example: Taking the question - what are vaccines? The metrics can be:

- Vaccines contain a weakened or deadened pathogen.
- The vaccine stimulates the immune system

- The immune system produces antibodies.

And so on.
If the teacher does not wish to manually input metrics, we can pre-compute and store metrics for a large database of questions generated using an LLM. This will require a SOTA model like Claude 4.7 or GPT 5.5 to ingest all questions once and generate metrics that can then be reused multiple times.

Once metrics are generated, the metrics and student answer can be fed into the crossencoder model in the following format:
[CLS] Student answer [SEP] Rubric [SEP]

Crossencoder models, part of traditional Natural Language Processing (NLP), use self-attention (as introduced by Attention is All You Need (Vaswani et al, 2017)) to analyze strong attention between matching terms and contextual attention between semantically related terms.
We will use an NLI (Natural Language Inference) crossencoder model rather than a generic Information Retrieval (IR) model to prevent false positives on highly similar but factually contradictory statements (e.g., confusing "oxygenated" with "deoxygenated").

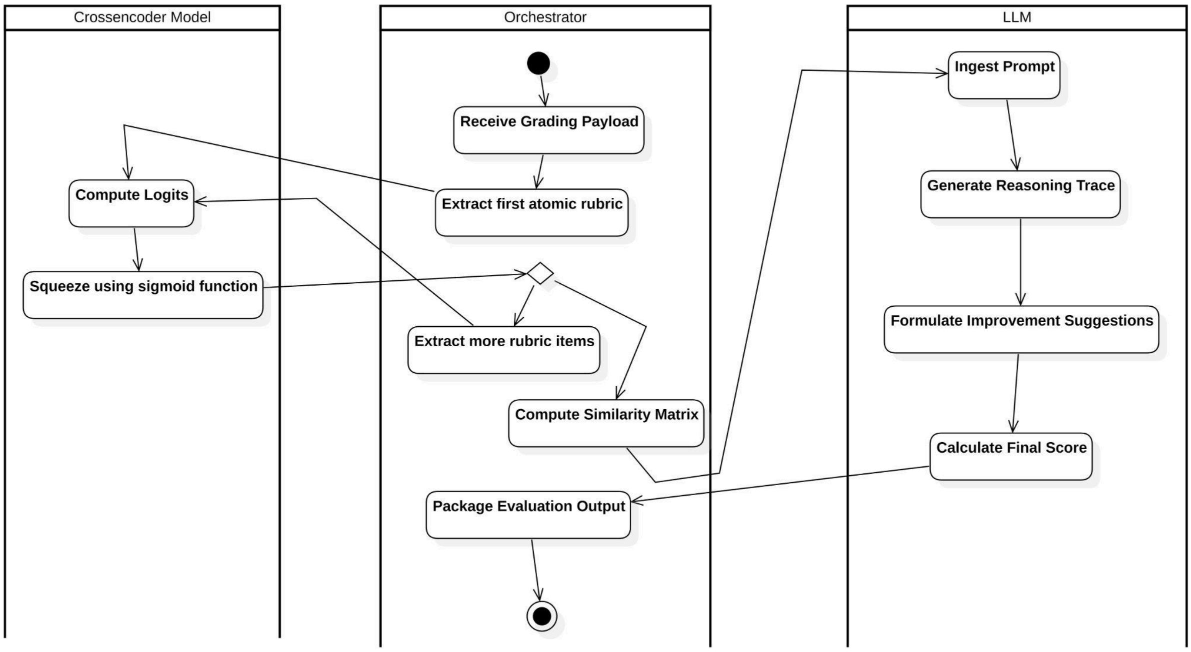
The output of crossencoder model is a relevance score, which is usually unbounded. We can then pass this output score into a sigmoid function to generate the final relevance score between 0 and 1.

The LLM prompt then takes in the relevance scores for each rubric, along with the original student answer as well as metadata like board, grade and maximum marks to generate improvement suggestions as well as a reasoning trace to finally assign a score to the answer.

By forcing the model to generate a reasoning trace and improvement suggestions, we eliminate the possibility of the model “randomly” assigning a score and then “hallucinating” justifications for the score.

Thus, this architecture allows us to use smaller LLMs for grading (as the crossencoding gives strong grounding signals to the LLM) which also reduces the cost.

Activity Diagram for Proposed Architecture



Evaluation of Proposed Architecture

To evaluate the proposed architecture, we need to focus first on the crossencoder.
We evaluate it like a standard binary classifier - entailment (matching) or no entailment (does not match).
Thus, we need to prepare a Golden Holdout Dataset of some answers where a known-good grader has matched the answer to the set of rubrics.
We can then calculate the precision, recall and F1 score of the crossencoder.

For evaluating the LLM, we turn to LLM-as-a-judge or Human-in-the-loop evaluation strategies. Here, we can measure trace adherence, feedback relevance and tone of the LLM output.

Finally, we can present the following as metrics of the system as a whole:

- Quadratic Weighted Kappa (QWK): This is the industry standard for automated essay/answer scoring. It measures the agreement between two raters (the AI and the human) while penalizing larger discrepancies more heavily than smaller ones (e.g., an AI grading a 4 instead of a 5 is a minor penalty; an AI grading a 1 instead of a 5 is a massive penalty).
- Exact Agreement Rate: What percentage of the time did the AI give the exact same total mark as the human grader?
- Adjacent Agreement Rate: What percentage of the time was the AI within 1 mark of the human grader?